NUMERICAL FAITHFULNESS AND FAILURE MODES IN FINANCIAL RAG SYSTEMS

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Abstract

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Retrieval-augmented generation (RAG) improves factual grounding by incorporating external evidence during generation, but financial question answering also requires accurate numerical reasoning. This thesis studies numerical faithfulness in financial RAG: whether generated numerical outputs are quantitatively correct, grounded in retrieved evidence, and consistent with the financial context.

We propose a lightweight evaluation framework combining numerical match metrics, relative error analysis, direct grounding measures, and heuristic error categorization. Using a FinQA-based retrieval-augmented generation setup over financial reports and structured tables, we evaluate lightweight RAG configurations under different retrieval depths, sparse and dense retrieval strategies, and preprocessing choices.

Results show that conventional text-based metrics can diverge substantially from numerical-faithfulness metrics. In particular, textual overlap and evidence grounding do not necessarily imply correct final numerical answers. We identify recurring failure patterns including abstention despite evidence, hallucinated numerical values, grounded but incorrect answers, and arithmetic or reasoning errors. The findings support numerical faithfulness as a distinct dimension of RAG evaluation and provide a practical basis for analyzing numerical reliability in financial AI systems.

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Use of AI-assisted tools. I used AI-assisted tools, namely the ChatGPT 5.5-Thinking model, during the preparation and revision of this thesis. These tools were used to support grammar correction, wording refinement, organization of revision notes, and limited drafting or rephrasing of explanatory passages. They were not used to conduct the experiments, generate the reported results, or replace my own analysis and interpretation. I reviewed, edited, and approved all text included in the final thesis. Approximately 20% of the thesis text was substantially revised with the assistance of these tools.

Finally, I am deeply grateful for the inspiration and encouragement drawn from the music of my favorite singer, Huang Xiaoyun. Her songs offered comfort, motivation, and emotional strength during many long and challenging stages of this work.

Contents

1 Introduction

1.1 Motivation

Modern economies are deeply shaped by the performance and governance of private and public corporations. Publicly traded companies collectively account for a substantial portion of national economic output, employment, and capital allocation. Ensuring transparency, accountability, and regulatory compliance in corporate reporting is therefore crucial for governments, investors, and regulatory bodies worldwide.

Regulatory standards in the financial sector are constantly evolving, and firms are required to disclose increasingly complex information through their annual reports, quarterly filings, and other regulatory documents. These documents often contain dense textual narratives, structured tables, and critical numerical indicators that inform investment decisions and compliance assessments. The growing volume and complexity of such disclosures have motivated the development of automated financial question answering (QA) systems capable of efficiently analyzing large collections of filings.

However, ensuring the reliability of these systems is paramount for accurate decision-making. Even minor numerical inaccuracies, misinterpretations of financial statements, or inconsistencies in reported figures can lead to substantial economic consequences, affecting investment strategies, risk assessments, and regulatory decisions [67]. Even as large language models (LLMs) are increasingly integrated into financial document analysis pipelines due to their impressive performance in natural language understanding and generation [74], they remain susceptible to important limitations that can compromise their outputs. One persistent challenge of LLMs is hallucination—the generation of factually incorrect or fabricated content presented with high confidence [44, 36]. In high-stakes financial settings, such hallucinations may manifest as incorrect numerical values, inconsistent calculations, or unsupported claims, thereby undermining trust in automated decision-support systems.

Retrieval-augmented generation (RAG) frameworks [52, 32] have emerged as a practical strategy to mitigate hallucination by grounding model outputs in external knowledge sources. Conventional RAG pipelines retrieve relevant document segments using semantic similarity between vector embeddings [45, 87], which are then incorporated into the generation process. Improvements in retrieval effectiveness have been driven by advances in embedding methods [66, 27], hybrid dense–sparse retrieval strategies, and hierarchical or late-interaction retrieval architectures [47, 90].

Specialized RAG systems have been proposed to handle the unique heterogeneity and regulatory complexity of financial filings. These systems include multi-modal retrieval pipelines

[86], agentic RAG systems leveraging Multiple Hypothetical Dynamic Embeddings (Multi-HyDE) [22, 26, 29], and domain-adapted retrieval strategies tailored to financial corpora [48]. These approaches collectively demonstrate substantial improvements in retrieval quality and answer-level performance for financial QA tasks.

Despite these advances, existing evaluation protocols for financial RAG systems primarily assess retrieval effectiveness (using metrics like recall, precision, F1, and Normalized Discounted Cumulative Gain) and overall answer quality (using text-based similarity or LLM-judged metrics such as ROUGE and RAGAS) [23]. While these metrics provide useful measures of a response's relevance and surface-level correctness, they do not explicitly disentangle retrieval quality from numerical reasoning fidelity, nor do they systematically evaluate whether generated numerical outputs are arithmetically consistent, properly grounded in retrieved evidence, or robust to retrieval variations. As a result, a system may demonstrate strong performance under these common evaluation metrics even when producing numerically incorrect answers.

Recent financial RAG studies have implicitly acknowledged this limitation. For example, [86] report lower satisfaction scores for financial status and performance queries, suggesting the need for more precise numerical retrieval and temporal reasoning. Similarly, [29] note that LLM-based evaluation may struggle to reliably assess primarily numeric answers and call for more comprehensive evaluation frameworks. These observations highlight a broader methodological gap: the lack of principled, automated metrics tailored to numerical faithfulness in generative financial QA systems.

This thesis addresses this methodological gap by proposing a structured evaluation framework for numerical faithfulness in financial RAG systems. Building on established financial QA benchmarks, we design automated metrics that assess arithmetic validity, numerical grounding, and the consistency of generated outputs. We then apply this framework to evaluate lightweight financial RAG configurations.

1.2 Problem Statement

This thesis considers the following setting in financial RAG: given a financial document corpus T and a user query Q, the system retrieves the top K relevant passages and generates an answer A conditioned on the retrieved evidence. While such systems aim to improve factual grounding, numerical inaccuracy remains a persistent challenge. This includes instances where the generated answer contains (i) numerical values that deviate from source documents, (ii) incorrect arithmetic reasoning or computations, or (iii) inconsistent or incorrect units.

To address this challenge, we introduce the concept of **numerical faithfulness**. Numerical faithfulness refers to the degree to which generated numerical outputs are arithmetically valid, properly grounded in retrieved evidence, and consistent with the financial context of the query. Unlike general factual correctness, numerical faithfulness requires precise verification of quantitative reasoning and numerical consistency.

Existing evaluation metrics do not explicitly measure numerical faithfulness. For example, traditional answer-level accuracy metrics often conflate retrieval effectiveness and reasoning correctness by evaluating only the final output. ROUGE-based metrics measure surface-level n-gram similarity and do not directly assess numerical correctness. Moreover, LLM-based evaluation has also been shown to be unreliable for primarily numeric responses. As an example, [29] observe that LLM judges may penalize minor rounding differences or, conversely, assign high scores despite substantive numerical errors. These inconsistencies suggest that current evaluation practices may obscure important quantitative failure modes in financial RAG systems.

A central difficulty is that an incorrect numerical answer can originate from different stages of the RAG pipeline. The retriever may fail to return the relevant financial evidence, the retrieved evidence may contain the correct numbers but lack table headers or units, or the generator may select the wrong operand, apply the wrong operation, or introduce an unsupported value. These cases can lead to the same outcome—an incorrect final answer—but they require different explanations and different remedies. Therefore, evaluating only final answer accuracy is insufficient for understanding numerical reliability in financial RAG systems.

Empirical evidence from financial QA benchmarks further highlights the problem of numerical faithfulness. FinanceBench reports substantial error and refusal rates even for advanced retrieval-augmented systems [39], while ConvFinQA emphasizes the difficulty of conversational financial queries requiring multi-step calculations [12]. These findings collectively indicate the need for a principled evaluation framework that explicitly targets numerical reliability in financial RAG systems.

1.3 Research Questions and Objectives

This thesis investigates how numerical faithfulness can be defined, measured, and analyzed in financial retrieval-augmented generation systems. Rather than evaluating financial RAG only through textual similarity or retrieval quality, the thesis focuses on whether generated numerical answers are correct, grounded in retrieved evidence, and arithmetically consistent.

The study is guided by the following research questions:

- **RQ1:** To what extent do the evaluated financial RAG configurations produce numerically faithful answers on FinQA [11]?
- **RQ2:** Can numerical errors be systematically identified, categorized, and measured using automatic metrics?

- **RQ3:** To what extent do conventional evaluation metrics fail to capture numerical inconsistencies?
- **RQ4:** Can lightweight verification mechanisms help assess numerical reliability in financial RAG systems?

To answer these questions, the thesis first formalizes numerical faithfulness as a combination of numerical accuracy, evidence grounding, and reasoning consistency. It then develops lightweight automatic metrics for evaluating these dimensions, applies them to a FinQA-based financial RAG setting, analyzes recurring numerical failure modes, and investigates a simple verification procedure for assessing percentage-change answers.

1.4 Contributions

This thesis makes the following three main contributions.

First, this thesis develops a numerical-faithfulness framework for evaluating financial RAG systems. Building on a focused review of RAG, information retrieval, grounded generation, RAG evaluation, and financial question answering, the thesis identifies a gap in existing evaluation protocols: standard metrics often measure retrieval quality, textual similarity, or final answer overlap, but do not explicitly indicate whether numerical answers are correct, grounded, and arithmetically valid. To address this gap, the thesis defines numerical faithfulness as a combination of three complementary conditions: the generated numerical answer should match the reference answer, the numerical values should be supported by retrieved evidence, and the answer should be consistent with valid arithmetic reasoning.

Second, this thesis proposes lightweight automatic measures for analyzing numerical faithfulness in financial RAG outputs. These include numerical match, relative error, direct grounding score, groundedness indicators, and conventional text-based metrics such as exact match and token F1 for comparison. Rather than relying on a single aggregate score, the evaluation reports these measures separately in order to distinguish answer correctness, evidence support, and numerical reliability. This separation helps show why a response can be textually similar or partially grounded while still being numerically incorrect.

Third, this thesis conducts an empirical study of lightweight financial RAG configurations on a FinQA-based benchmark setting and analyzes their numerical failure modes. The experiments compare retrieval depths, retrieval strategies, and preprocessing choices under both conventional and numerical-faithfulness metrics. The analysis shows that conventional text-based metrics, retrieval grounding, and numerical correctness can diverge. To interpret these patterns, the thesis proposes a heuristic failure taxonomy covering retrieval-induced errors, arithmetic or reasoning errors, substitution errors, unit and scaling errors, hallucinated numerical values, and abstention despite evidence. In addition, the thesis investigates a lightweight post-generation verification strategy for percentage-change questions, showing how simple verification can act as a warning layer for a subset of numerical reliability issues.

1.5 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews background and related work on large language models, information retrieval, retrieval-augmented generation, retrieval and ranking techniques, grounded generation, RAG evaluation, financial RAG systems, and financial QA benchmarks. Chapter 3 presents the methodology, including the financial RAG task definition, the formal definition of numerical faithfulness, the numerical error taxonomy, the FinQA-based RAG setup, and the evaluation metrics. Chapter 4 presents the experiments and analysis, including baseline performance, preprocessing and retrieval comparisons, numerical failure modes, dense retrieval comparison, and lightweight verification strategies. Chapter 5 concludes the thesis by summarizing the main contributions, discussing implications and limitations, and outlining future directions.

2 Background and Related Work

This chapter reviews prior work that forms the foundation of retrieval-augmented generation (RAG) systems for financial question answering. We first discuss the limitations of parametric knowledge in large language models and explain the need for external retrieval mechanisms. We then summarize classical and neural information retrieval techniques that underpin modern RAG pipelines. Next, we examine representative RAG architectures and extensions, including advances in retrieval, ranking, grounded generation, and RAG evaluation. Finally, we review financial RAG systems and financial QA benchmarks, with emphasis on how existing work motivates the need to evaluate numerical correctness, grounding, and reliability in financial RAG outputs.

2.1 Large Language Models and Parametric Knowledge

Modern large language models are built on the Transformer architecture, introduced by Vaswani et al. [82]. Later models such as GPT-2 and BERT showed two important directions for large-scale pretraining: autoregressive language modeling for text generation and masked language modeling for bidirectional representation learning [21, 65]. Subsequent model families, including GPT, PaLM, and LLaMA, further demonstrated that scaling Transformer-based models can improve in-context learning, reasoning, and language understanding [7, 15, 61, 80, 81].

A key property of LLMs is that they store knowledge parametrically. Instead of maintaining explicit facts in a database, pretrained models encode factual associations implicitly in their weights. Prior work shows that language models can recover relational and factual knowledge from prompts, and that larger models can answer many factual questions without external retrieval [62, 69]. However, this knowledge is distributed across the model and is difficult to inspect, update, or remove reliably [4, 19, 68, 83].

This creates important reliability problems. LLMs may generate hallucinated content, meaning fluent but unsupported or factually incorrect statements [44, 58]. They also have limited temporal coverage because their internal knowledge depends on the data available during training. Even reported knowledge cutoff dates may not fully describe what a model actually knows, since training corpora can contain duplicated, outdated, or temporally misaligned data [13].

These limitations motivate the distinction between parametric and non-parametric memory. Parametric memory is stored inside model weights and supports fluent generalization, but it is opaque and difficult to verify. Non-parametric memory, such as documents, databases, and retrieval indexes, is external, inspectable, and easier to update. This distinction provides the motivation for retrieval-augmented generation, where language models are connected to external evidence rather than relying only on internal knowledge.

2.2 Information Retrieval Foundations

Information retrieval (IR) provides the retrieval foundation for RAG systems. Its goal is to find and rank documents that are relevant to a user query. Classical IR methods usually represent documents and queries using sparse lexical features, such as bag-of-words or TF-IDF [72, 73]. These methods ignore much of the sentence structure, but they are efficient, interpretable, and effective for exact keyword matching.

BM25 is a widely used sparse retrieval method that improves on simple term matching by accounting for term frequency, inverse document frequency, and document length normalization [70]. In large corpora, sparse retrieval is typically implemented with inverted indexes, which allow systems to quickly find documents containing the query terms [57]. Although these methods may struggle with paraphrases or semantic similarity, they remain strong baselines for RAG because they are efficient and reliable when exact terms, entities, or numbers matter.

Dense retrieval extends classical lexical search by representing queries and documents as learned semantic vectors. Early methods such as Latent Semantic Analysis (LSA) and topic models map text into lower-dimensional semantic spaces, while neural models such as Deep Structured Semantic Model (DSSM) learn continuous representations for semantic matching [6, 20, 33, 37]. Later work emphasized that retrieval requires not only general semantic similarity but also fine-grained relevance matching between queries and documents [31]. Modern dense retrievers, such as Dense Passage Retrieval (DPR), use dual encoders to embed queries and passages into a shared vector space and rank passages by vector similarity [45]. Further improvements such as RocketQA and Approximate Nearest Neighbor Negative Contrastive Estimation (ANCE) strengthen dense retriever training through better negative sampling and hard-negative mining [64, 87]. Overall, dense retrieval helps RAG systems retrieve semantically related evidence beyond exact keyword overlap, although sparse methods remain useful when exact terms, entities, years, or numbers are important.

Modern retrieval systems often use hybrid and multi-stage designs. A first-stage retriever, such as BM25 or a dense retriever, collects a candidate set, and a second-stage reranker scores those candidates more carefully. Cross-encoder rerankers jointly encode the query and passage, allowing fine-grained relevance scoring, but they are more expensive because each query–passage pair must be processed separately [60].

Late-interaction models such as ColBERT provide another trade-off between accuracy and efficiency. ColBERT represents queries and documents with contextualized token embeddings and compares them using token-level matching, while still allowing document representations to be precomputed [47]. Together, cross-encoder reranking, late interaction, and hybrid sparse-dense retrieval show that retrieval quality depends not only on finding relevant passages, but also on ranking evidence in a way that supports downstream generation.

2.3 Retrieval-Augmented Generation (RAG)

Lewis et al. [52] introduce RAG as a framework that combines a parametric sequence-to-sequence generator with a non-parametric retrieval memory. Given an input query, the model retrieves relevant passages from an external corpus and conditions generation on both the query and the retrieved evidence. The paper studies two variants: RAG-Sequence, which uses the same retrieved passage for the whole output, and RAG-Token, which can use different retrieved passages during token-level generation.

Related early work also explored tighter integration between retrieval and language modeling. Guu et al. [32] propose REALM, which incorporates a learned retriever into language model pre-training, allowing the model to retrieve documents during pre-training, fine-tuning, and inference. Izacard et al. [42] later introduce Atlas, a retrieval-augmented language model designed for knowledge-intensive tasks with limited supervision, showing that retrieval can improve few-shot performance when the retriever and generator are trained together.

These foundational systems show that retrieval can improve language models by providing external, inspectable, and updateable knowledge. However, they also motivate later concerns studied in this thesis: retrieving relevant evidence does not guarantee that the generated answer is faithful, numerically correct, or consistent with the required reasoning process. This limitation is especially important in financial RAG, where the model must not only retrieve relevant passages but also select the correct values, preserve units and reporting periods, and perform valid calculations.

2.4 Retrieval and Ranking Techniques in RAG

The effectiveness of a RAG system depends not only on the generator, but also on how evidence is retrieved and ranked. Retrieval design choices such as document chunking, overlapping windows, metadata filtering, sparse-dense hybrid search, and hierarchical indexing can strongly affect the quality of the context given to the model [28].

These choices matter because retrieval is not only about finding many potentially relevant passages. The retrieved evidence must also be useful, specific, and appropriate for the task. In financial RAG, this is especially important because nearby passages may contain similar numbers, different reporting periods, or different units. Poor retrieval can therefore lead to incorrect numerical answers even when the model appears to be grounded.

Chunking is an important preprocessing step in RAG because long documents must be divided into retrievable units before indexing [79]. Larger chunks can preserve more context, but they may also include irrelevant information; smaller chunks can improve retrieval precision, but they may remove context needed for interpretation. To reduce fragmentation, systems often use recursive splitting, overlapping windows, or hierarchical retrieval, where small chunks are indexed first and then expanded to larger parent sections when selected [9, 50]. These choices are especially important in financial RAG because table headers, reporting periods, units, and nearby values may be separated if the chunking strategy is too narrow.

Metadata-aware retrieval uses structured signals such as dates, section headers, entities, or document-level summaries to improve retrieval and ranking. Instead of treating each chunk as only plain text, these methods add extra information that can help the retriever distinguish between similar passages. This is especially useful in structured domains, where relevant evidence may depend on document type, reporting period, or section context [28].

Prior work has explored several ways to incorporate metadata into retrieval, including adding metadata to chunk text, combining metadata and content embeddings, and using metadata for filtering or reranking [88]. In financial RAG, metadata-driven approaches have also been used to enrich document and chunk representations with summaries, entities, and anticipated question-answer signals [17]. These studies suggest that metadata can improve retrieval quality by making domain-specific context more explicit.

Hybrid retrieval combines sparse and dense retrieval so that a RAG system can benefit from both exact term matching and semantic similarity. Sparse methods such as BM25 and TF-IDF are useful for matching specific keywords, while dense retrievers can capture paraphrases and broader semantic relationships [41, 45, 70, 72].

Recent work has used hybrid retrieval to improve retrieval quality and reduce hallucination in RAG. For example, Mala et al. [56] combine sparse and dense retrievers using Reciprocal Rank Fusion rather than direct score interpolation [8, 16]. Their approach also adjusts the balance between sparse and dense retrieval depending on query type. This is relevant to financial RAG because some questions require exact matching of terms, years, or financial variables, while others require broader semantic matching across related evidence.

Huang et al. [35] propose HiRAG, a graph-based RAG method that organizes retrieved knowledge across multiple levels rather than using only a flat graph. Its indexing stage builds higher-level summary entities from lower-level entity clusters, while its retrieval stage combines local entity information, global community summaries, and bridge-level reasoning paths. This is relevant to RAG because it shows that retrieval can benefit from preserving relationships between fine-grained evidence and broader document-level context.

These retrieval and ranking techniques show that retrieval is not just a preprocessing step in RAG. Choices such as chunking, metadata use, hybrid ranking, and hierarchical indexing shape the evidence that the generator receives. However, stronger retrieval does not automatically lead to faithful generation, especially when retrieved passages contain redundant, conflicting, or task-irrelevant information.

For financial RAG, retrieval must therefore be evaluated not only by top-K recall, but also by whether the retrieved evidence supports correct numerical reasoning. The next section discusses how RAG systems attempt to improve grounding and faithfulness after evidence has been retrieved.

2.5 Grounded and Faithful Generation

RAG systems use retrieved evidence to improve the factual reliability of generated answers. However, retrieving relevant evidence does not guarantee that the final answer is faithful to that evidence. Prior work on grounded generation, citation-aware RAG, and faithfulness evaluation shows that models may retrieve useful passages but still generate unsupported claims, use evidence incorrectly, or attach citations that do not fully justify the answer [3, 24, 40, 84].

In this thesis, grounding refers to the connection between generated content and retrieved evidence, while faithfulness refers to whether the answer is actually supported by that evidence. This distinction is important for financial RAG because a generated number may appear in the retrieved context but still be selected, interpreted, or calculated incorrectly.

This section reviews work on grounded generation, citation-aware RAG, self-reflective retrieval, and faithfulness evaluation. These studies motivate the need to evaluate not only whether evidence is retrieved, but also whether the generated answer uses that evidence correctly.

Asai et al. [3] and Islam et al. [40] both study retrieval-aware generation methods that move beyond static retrieve-then-generate pipelines. SELF-RAG trains a model to decide when retrieval is needed and to evaluate retrieved passages and generated outputs using reflection tokens. OPEN-RAG builds on a similar idea but focuses more on reasoning with open-source models, including adaptive retrieval, support assessment, and training against misleading distractor evidence. Together, these works suggest that reliable RAG may require not only retrieving evidence, but also deciding when to retrieve, judging whether evidence is relevant, and checking whether the generated answer is supported by that evidence.

Fadeeva et al. [24] introduce FRANQ, a framework for uncertainty estimation in RAG that separates factuality from faithfulness to retrieved context. The paper argues that a claim can be factually correct even if it is not supported by the retrieved passages, and that a claim can be faithful to retrieved evidence while still being factually wrong if the retrieved evidence is misleading. This distinction is relevant to this thesis because numerical faithfulness also requires separating surface grounding from actual correctness: a generated financial number may appear in the retrieved evidence but still be incorrect, or may be correct without being properly grounded in the retrieved context.

Wallat et al. [84] distinguish citation correctness from citation faithfulness in RAG systems. A citation may be correct if the cited document supports the generated statement, but it may still be unfaithful if the model did not actually rely on that document when producing the answer. Their study shows that models can sometimes attach citations after the fact, a behavior they describe as post-rationalization. This is relevant to this thesis because it reinforces the distinction between surface evidence support and faithful evidence use: a financial answer may cite or retrieve a relevant number, but still fail to use the evidence in a faithful numerical reasoning process.

Abolghasemi et al. [1] study attribution bias in generator-aware RAG systems. Their work shows that adding authorship metadata to retrieved documents can change how models cite sources, and that evaluated models tend to favor documents labeled as human-written. This is relevant to RAG faithfulness because it shows that attribution is not purely evidence-based: metadata and presentation cues can influence which sources a model trusts and cites.

Qian et al. [63] propose VeriCite, a citation-aware RAG framework that verifies and refines evidence before producing the final answer. The system first generates an initial response, checks whether its claims are supported by retrieved passages, selects useful supporting evidence, and then rewrites the verified content into a final cited answer. Although VeriCite is not specific to financial QA, it is relevant because it shows that attribution requires more than attaching citations to generated text; the supporting evidence itself must be checked before the answer can be considered faithful.

Hu et al. [34] propose CG-RAG, a graph-based RAG framework for research question answering over citation networks. Instead of treating retrieved passages as independent chunks, CG-RAG uses citation graph structure to connect related document segments and combines lexical, semantic, and graph-based retrieval signals. Although this work focuses on scientific literature rather than financial documents, it is relevant because it shows that retrieval quality can improve when document structure and inter-document relationships are considered during evidence selection.

Asai et al. [2] introduce OpenScholar, a retrieval-augmented system for scientific literature synthesis. OpenScholar combines a scientific document store, trained retrievers and rerankers, citation-backed generation, and a self-feedback loop for refining long-form answers. Although it targets scientific literature rather than financial QA, it is relevant because it shows that retrieval must be paired with attribution and verification mechanisms to support faithful generation.

Recent work on citation- and attribution-aware generation shows that high-recall retrieval is not enough to ensure faithful RAG outputs. Even when relevant passages are retrieved, the model may still generate claims that are weakly supported, incorrectly attributed, or not directly grounded in the evidence.

These studies support an important point for this thesis: retrieval does not automatically guarantee grounding, and grounding does not automatically guarantee numerical correctness. Financial RAG systems therefore require evaluation methods that check not only whether evidence is retrieved, but also whether the generated answer uses that evidence correctly.

2.6 Evaluation of RAG Systems

Evaluating RAG systems is more complex than evaluating standalone language models because both retrieval and generation affect the final answer. A system may retrieve incomplete evidence, retrieve relevant but noisy evidence, or generate an answer that is fluent but not fully supported by the retrieved context. Therefore, RAG evaluation must consider not only answer quality, but also evidence relevance, grounding, and attribution.

Existing evaluation practices often measure these aspects separately. Retrieval metrics assess whether relevant passages are selected, while generation metrics assess surface-level answer quality or textual overlap. However, these metrics may not reveal whether the generated answer uses the retrieved evidence correctly. This limitation is especially important in financial RAG, where an answer can appear grounded while still containing an incorrect number or invalid calculation.

This thesis addresses on this gap by focusing on numerical faithfulness. Rather than treating RAG reliability as a single score, the proposed evaluation separates numerical correctness, evidence grounding, and reasoning consistency.

Historically, RAG evaluation metrics have depended on the target task. For example, QA tasks typically use exact-match (EM) and F1-scores [25, 38, 77, 85], whereas accuracy is the primary evaluation metric of fact-checking tasks [75]. Answer quality is often evaluated via BLEU and ROUGE [5, 30, 46, 76].

RAG evaluation is commonly separated into retrieval-side and generation-side assessment. Retrieval metrics, such as Hit Rate, MRR, and NDCG, measure whether useful evidence is retrieved and ranked highly [53, 59]. Generation metrics evaluate whether the produced answer is relevant, coherent, faithful to the retrieved evidence, and factually reliable [49, 51].

Recent RAG evaluation frameworks also examine whether systems retrieve relevant context, generate answers supported by that context, and respond directly to the user query [23, 43, 71]. Other work evaluates system robustness, including whether a model can handle noisy retrieval results, reject unanswerable queries, ignore misleading evidence, and combine information from multiple documents [10, 54].

Several specialized tools and systems have been developed to evaluate these dimensions of RAG systems. These include benchmarks such as Retrieval-Augmented Generation Benchmark (RGB) [10] and Create-Read-Update-Delete (CRUD) [55], which focus on evaluating the essential capabilities of RAG models. Automated evaluation tools such as RAGAS [23], ARES [71], and

TruLens [18] use LLM-based or model-assisted methods to score different dimensions of RAG performance.

Together, these tools and benchmarks provide useful methods for evaluating RAG systems, but they also highlight the limitations of current evaluation practices. As RAG systems expand beyond QA tasks and are increasingly deployed in decision-support tasks and complex, multi-stage applications, evaluation metrics that capture both retrieval quality and generation reliability become increasingly important.

2.7 Financial RAG Systems

This section reviews recent financial RAG systems that are closely related to the setting of this thesis. These systems address challenges such as complex financial filings, heterogeneous text-table evidence, domain-specific retrieval, metadata enrichment, and multi-step financial question answering.

Wang et al. [86] propose FinSage, a financial RAG framework for question answering over complex financial filings. The paper focuses on the difficulty of applying RAG to documents that contain heterogeneous information, including narrative text, tables, figures, and metadata. To address this issue, FinSage combines document preprocessing, multi-path retrieval, and domain-specific reranking. The preprocessing stage converts different document formats into machine-readable chunks with additional metadata; the retrieval stage combines sparse, dense, metadata-based, and HyDE-style retrieval signals; and the reranking stage is trained to prioritize financial evidence that is more relevant to compliance-oriented questions.

Similar to FinSage, Srinivasan et al. [29] study financial RAG in the setting of complex financial filings and multi-step financial questions. Their system combines retrieval with an agentic workflow, allowing the model to decompose queries, select tools, refine intermediate results, and generate a final answer from accumulated evidence. A central part of their approach is Multi-HyDE, which extends standard HyDE by generating multiple related but non-equivalent queries and then producing a hypothetical document for each query. This design aims to broaden retrieval coverage while preserving the precision needed for financial documents, where similar passages may differ in important numerical or temporal details.

Kim et al. [48] approach financial RAG from the perspective of retrieval optimization. Their pipeline organizes the process into three stages: pre-retrieval, retrieval, and post-retrieval refinement. In the pre-retrieval stage, they examine ways to improve both the user query and the financial corpus, including query expansion and document restructuring. In the retrieval stage, they adapt embedding models to financial text and combine dense retrieval with sparse retrieval so that the system can capture both semantic similarity and exact financial terminology. In the post-retrieval stage, they apply reranking and document selection to reduce irrelevant context before answer generation.

Dadopoulos et al. [17] study metadata-driven RAG for financial question answering. Their work focuses on long and structured financial filings, where relevant evidence may be sparse, distributed across sections, and difficult to retrieve with a flat chunk-based index. The proposed framework enriches financial document chunks with LLM-generated metadata and evaluates three main types of retrieval enhancement: pre-retrieval optimization, post-retrieval refinement, and semantic embedding enrichment. Pre-retrieval optimization uses document-level metadata to filter files and rewrite queries before retrieval. Post-retrieval refinement uses metadata to rerank retrieved chunks and, in some settings, to expand the retrieved context through related entities and clusters. Semantic embedding enrichment incorporates chunk-level metadata directly into the text representation before embedding, producing contextual chunks that better preserve the structure and meaning of financial reports.

Together, these studies show that financial RAG systems often require domain-aware retrieval, careful preprocessing, reranking, metadata use, and support for complex financial evidence. However, their evaluations commonly emphasize retrieval effectiveness, answer quality, or system-level performance. This thesis builds on this line of work by focusing specifically on numerical faithfulness: whether generated financial answers are numerically correct, grounded in retrieved evidence, and consistent with the required calculation.

2.8 Financial QA Benchmarks

Financial QA benchmarks provide controlled settings for evaluating whether models can retrieve, interpret, and reason over financial evidence. They are especially relevant to numerical-faithfulness evaluation because many financial questions require table understanding, scale interpretation, evidence selection, and arithmetic reasoning. This section summarizes the benchmarks most closely related to the FinQA-based setting used in this thesis.

Chen et al. [11] introduce FinQA, a financial question-answering dataset designed for numerical reasoning over company reports. The dataset includes questions over textual evidence and structured tables, together with supporting facts, final answers, and structured reasoning programs. FinQA is suitable for this thesis because its questions require systems to select relevant financial values and perform numerical reasoning. In this thesis, FinQA is reformulated as a RAG task: the model is not given the annotated supporting evidence or reasoning program during generation, but must retrieve relevant evidence from the financial document and generate an answer from the retrieved context.

Chen et al. [12] extend FinQA into a conversational financial QA setting through ConvFinQA.

The dataset focuses on sequential questions over financial reports, where later questions may depend on earlier turns. ConvFinQA is related to this thesis because it highlights how numerical errors can accumulate across turns. However, this thesis focuses on single-turn financial RAG over retrieved evidence rather than conversational QA.

Zhu et al. [89] introduce TAT-QA, a financial question-answering benchmark built around hybrid contexts that combine tables with related text. Many questions require the model to connect information across these two sources and perform numerical reasoning, such as comparison, arithmetic, or multi-step composition. TAT-QA is relevant to this thesis because it emphasizes evidence selection, scale interpretation, and answer derivation in financial text-table reasoning.

Choi et al. [14] introduce FinDER, a financial question-answering benchmark designed for retrieval-augmented settings. Unlike benchmarks that provide a fixed context with each question, FinDER requires systems to retrieve relevant evidence from financial documents before generating an answer. This makes it useful for evaluating financial RAG systems because it tests both retrieval quality and answer generation under realistic financial queries.

Strich et al. [78] introduce T^2-RAGBench, a text-and-table benchmark for evaluating financial RAG systems. It is designed for an unknown-context setting, where the system must retrieve the relevant text-and-table evidence before answering. A key feature is its use of context-independent questions, which are intended to be answerable without relying on an oracle-provided passage. This makes the benchmark especially relevant to RAG evaluation because it tests whether the system can retrieve the correct evidence rather than simply answer from a predefined context.

Together, these benchmarks show that financial QA often requires evidence selection, table understanding, scale interpretation, and numerical reasoning. FinQA is used in this thesis because it provides financial questions, structured tables, final answers, and reasoning annotations that are well aligned with the study of numerical faithfulness. At the same time, related benchmarks such as ConvFinQA, TAT-QA, FinDER, and T^2-RAGBench show that numerical faithfulness is relevant beyond the specific FinQA-based setup evaluated here.

These benchmarks are not evaluated directly in this thesis because the experimental scope is limited to a controlled FinQA-based RAG setup; broader evaluation on ConvFinQA, TAT-QA, FinDER, and T2-RAGBench is left for future work.

3 Methodology

3.1 Financial RAG Task Definition

In this thesis, financial retrieval-augmented generation (RAG) refers to a system that answers questions about financial documents by first retrieving relevant evidence and then generating an answer from that evidence.

Let $\mathcal{D} = \{d_1, d_2, \ldots, d_n\}$ be a corpus of financial documents. Given a query Q, the retriever selects a set of relevant passages $\mathcal{R} = \{r_1, r_2, \ldots, r_K\}$. The language model then generates an answer conditioned on the query and the retrieved passages:

$$A = \text{LM}(Q, \mathcal{R}). \tag{1}$$

Financial RAG is especially challenging because financial reports contain dense numerical information, tables, reporting periods, units, and repeated values with different meanings. A model may retrieve relevant evidence but still produce an incorrect answer by selecting the wrong number, using the wrong year, applying the wrong operation, or introducing an unsupported value.

For this reason, this thesis focuses on numerical faithfulness: whether a generated numerical answer is correct, supported by retrieved evidence, and consistent with the required calculation.

3.2 Formal Definition of Numerical Faithfulness

In this section, we formalize the notion of numerical faithfulness (NF), which represents the degree to which generated numerical outputs remain consistent with retrieved evidence and underlying financial data.

Let Q denote a user query, $\mathcal{D}$ a document corpus, $\mathcal{R} \subset \mathcal{D}$ the set of retrieved passages, A the generated answer, and A^* the ground truth answer. We define $\mathcal{N}(X)$ as the set of numerical values extracted from text X. In this case, $\mathcal{N}(\mathcal{R})$, $\mathcal{N}(A)$, and $\mathcal{N}(A^*)$ denote the sets of numbers appearing in the retrieved evidence, generated answer, and ground truth answer, respectively.

We define numerical faithfulness as the extent to which the numerical content of a generated answer is correct, grounded in the retrieved evidence, and derived through logical reasoning steps. Formally, numerical faithfulness is satisfied if the following three conditions hold:

$$\text{NF}(A, A^*, \mathcal{R}) = 1 \quad \text{if all conditions below are satisfied.} \tag{2}$$

(1) Numerical Accuracy. The numerical value reported in the final-answer field should be consistent with the ground truth answer. Let $N(A_{\text{final}})$ denote the set of numerical values extracted from the final-answer field of the generated output. For each predicted final-answer number $n \in N(A_{\text{final}})$, there should exist a corresponding ground-truth number $n^* \in N(A^*)$ such that n and n^* are approximately equal under a tolerance function, such as relative error or scale-invariant comparison. This condition focuses on the final reported answer rather than all numbers appearing in the explanation, since explanations may contain years, operands, units, or intermediate calculation values that are not intended to match the final gold answer.

(2) Numerical Grounding. The generated numerical values should be supported by the retrieved evidence. For each $n \in \mathcal{N}(A)$, there exists some passage $r \in \mathcal{R}$ such that n appears in r or can be validly derived from the numerical values in $\mathcal{N}(\mathcal{R})$. This condition ensures that the model does not introduce unsupported or hallucinated numerical values.

(3) Numerical Reasoning Consistency. When the answer requires computation, the generated numerical values should be derived through valid arithmetic reasoning. Such reasoning includes mathematical operations such as summation, subtraction, ratio computation, or percentage change. The computed results should be consistent with the underlying numerical evidence and the intended operation specified by the query.

Together, these conditions define numerical faithfulness as a combination of answer correctness, evidence support, and calculation validity. They are complementary rather than interchangeable. A generated answer may match the ground truth while not being supported by the retrieved evidence, or it may use numbers from the retrieved context while applying an incorrect operation. In this thesis, this definition is used as a conceptual criterion rather than as a single binary metric computed directly in the experiments. The empirical evaluation instead approximates these dimensions through separate measures, including numerical match, relative error, grounding scores, and lightweight verification flags. Future work could extend this framework into a continuous numerical-faithfulness score by combining these dimensions with validated weights.

3.3 Dataset and RAG Setup

This thesis uses FinQA [11] as the primary benchmark for evaluating numerical faithfulness in financial RAG. FinQA is selected because it contains financial questions, structured tables, reference answers, supporting evidence, and program-style reasoning annotations.

FinQA is well aligned with the goal of this thesis because many of its questions require numerical reasoning over financial text and tables. In addition to final answers, FinQA provides gold supporting evidence and gold reasoning programs[1], which make it possible to distinguish answer

[1] In this thesis, "gold" refers to the reference annotations provided by the benchmark dataset, such as the gold answer, gold supporting evidence, or gold reasoning program.

correctness from the reasoning process that produces the answer. Although this thesis focuses mainly on answer-level numerical faithfulness, grounding, and heuristic failure analysis, these annotations make FinQA a suitable foundation for future program-level verification.

This thesis reformulates FinQA from its original benchmark setting into a retrieval-augmented generation setting. Instead of providing the gold supporting context or gold reasoning program to the model during generation, the system must retrieve relevant passages from the financial document and then generate an answer from the retrieved evidence. This allows numerical faithfulness to be evaluated under retrieval uncertainty.

For each example, the original financial document is treated as the retrieval corpus. The document is segmented into candidate passages, indexed by the retrieval method, and queried using the corresponding financial question. The generator receives the question and the retrieved top-K passages in a fixed prompt format.

The experimental pipeline follows a simple two-stage RAG design. The retriever first selects the top-K passages, and the generator then produces an answer conditioned on those passages. For example, the primary retriever used in this thesis is BM25, with additional lightweight retrieval variants such as keyword overlap and optional year-overlap scoring. These methods are intentionally simple and interpretable so that retrieval behavior can be analyzed in relation to numerical errors.

For generation, this thesis uses a locally served instruction-following model through Ollama, with Qwen2.5 as the main generator. The prompt instructs the model to answer using only the provided evidence, preserve units carefully, write required calculations explicitly, and abstain when the evidence is insufficient.

The same prompt template is used across all retrieval configurations so that differences in output can be attributed primarily to retrieval depth, retrieval strategy, and preprocessing choices rather than prompt changes. The main prompt template is shown below.

```
You are answering a financial question using only the provided evidence.

Question:
{question}

Evidence:
{context}

Instructions:
- Use only the evidence above.
- If calculation is needed, write the equation explicitly.
```

```
- Preserve units carefully.
- Do not use outside knowledge.
- If the evidence is insufficient, write exactly:
  EQUATION: Insufficient evidence
  FINAL ANSWER: Insufficient evidence

Required output format:
EQUATION: <equation or brief reasoning>
FINAL ANSWER: <final numeric answer with unit if needed>

Now answer.
```

All experiments are implemented in Python using a unified retrieval, generation, and evaluation pipeline. The pipeline records the question, gold answer, retrieved passages, retrieval configuration, model name, and generated output for each example. Generated outputs are saved as JSONL files and evaluated with both conventional answer-level metrics and numerical-faithfulness metrics.

3.4 Taxonomy of Numerical Errors

To analyze numerical-faithfulness failures more precisely, this thesis uses a taxonomy of common numerical error types in financial RAG outputs. These categories describe different ways in which a generated numerical answer may fail: the retrieved evidence may be incomplete, the model may select the wrong value, the calculation may be invalid, the unit or scale may be misread, or the output may contain a number unsupported by the retrieved context. Illustrative examples are provided after Table 1 to clarify how these categories differ from one another.

Table 1: Taxonomy of numerical error types in financial RAG systems.

Error Type	Description
Arithmetic Error	The numerical result is incorrect due to invalid computation, such as incorrect aggregation, ratio calculation, or arithmetic operations.
Unit Mismatch	The numerical value is expressed in an incorrect unit or scale (e.g., millions vs. billions), leading to inconsistency in magnitudes.
Substitution Error	Incorrect numerical values are used in place of the correct ones, often due to selecting the wrong figures from retrieved evidence.
Hallucinated Number	The generated number does not appear in the retrieved evidence and cannot be derived from it, indicating unsupported generation.
Retrieval Error	The generated number is grounded in the retrieved evidence but differs from the ground truth due to incorrect or incomplete retrieval.
Abstention Despite Evidence	The system refuses to answer or states that there is insufficient evidence even though the retrieved context contains enough information to compute or support the answer.

3.4.1 Arithmetic Error

An arithmetic error occurs when the retrieved evidence contains the correct numerical operands, but the generated answer applies an incorrect computation or reports an incorrect numerical result. For example, consider the question:

> What is the percentage change in revenue from 2010 to 2011?

Suppose the retrieved evidence contains the correct revenue values:

- 2010 Revenue = $63,003
- 2011 Revenue = $23,596

The correct computation is:

$$\frac{23{,}596 - 63{,}003}{63{,}003} \times 100 = -62.49\%. \tag{3}$$

If the RAG system instead outputs -52%, then the relevant evidence was retrieved, but the numerical result is incorrect. This failure is therefore not primarily a retrieval failure. Rather, it reflects an arithmetic or answer-generation error because the model had access to the correct operands but produced an incorrect final value.

3.4.2 Unit Mismatch

A unit mismatch occurs when the generated answer uses the correct general quantity but expresses it using an incorrect unit or scale. For example, suppose a financial table states that all values are reported in millions of dollars, and the retrieved evidence contains the following value:

- Revenue = 23.6, with table unit: millions of dollars

The correct interpretation is therefore:

$$23.6 \text{ million dollars} = 23{,}600{,}000 \text{ dollars.} \tag{4}$$

However, if the generated answer interprets the value as \$23.6 billion, then the answer is wrong by a factor of 1,000:

$$23.6 \text{ billion dollars} = 23{,}600{,}000{,}000 \text{ dollars.} \tag{5}$$

In this case, the generated number may appear to be grounded in the retrieved evidence because it uses the value 23.6, but its magnitude is inconsistent with the unit specified by the table header, caption, or surrounding text. This type of error is important in financial RAG because units are often stated once at the table level rather than repeated in every cell.

3.4.3 Substitution Error

A substitution error occurs when the model retrieves relevant evidence but selects the wrong numerical value from that evidence. For example, suppose the question asks:

> What is the percentage change in revenue from 2010 to 2011?

The correct evidence is:

- 2010 Revenue = \$63,003
- 2011 Revenue = \$23,596

The correct computation is:

$$\frac{23{,}596 - 63{,}003}{63{,}003} \times 100 = -62.49\%. \tag{6}$$

However, suppose the retrieved table also contains another nearby financial value:

- 2011 Operating Income = \$35,000

If the model incorrectly substitutes operating income for 2011 revenue, it may compute:

$$\frac{35{,}000 - 63{,}003}{63{,}003} \times 100 = -44.45\%. \tag{7}$$

This differs from an arithmetic error because the computation itself is valid, but the operands are wrong. The failure comes from substituting an incorrect number for the required one. In financial documents, this can occur when tables contain many adjacent rows, years, and related financial quantities with similar formatting.

3.4.4 Hallucinated Number

A hallucinated number occurs when the generated answer contains a numerical value that is neither present in the retrieved evidence nor derivable from the retrieved numbers. For example, suppose the retrieved context contains only the following values:

- 2010 Revenue = $63,003
- 2011 Revenue = $23,596

The correct percentage change can be derived from these values:

$$\frac{23{,}596 - 63{,}003}{63{,}003} \times 100 = -62.49\%. \tag{8}$$

However, suppose the model generates the following explanation:

> The revenue decreased from $63,003 to $30,247, resulting in a decrease of approximately 52%.

The value $30,247 does not appear in the retrieved evidence and is not derivable from the two retrieved revenue values. In this case, the answer fails the grounding requirement because the generated number lacks evidential support. This type of failure is especially problematic in financial settings because unsupported numerical values may appear precise even when they are not justified by the source document.

3.4.5 Retrieval Error

A retrieval error occurs when the generator answers based on retrieved evidence that is incomplete, irrelevant, or associated with the wrong financial context. For example, suppose the question asks:

> What is the percentage change in revenue from 2010 to 2011?

The correct evidence should contain:

- 2010 Revenue = \$63,003
- 2011 Revenue = \$23,596

The correct computation would be:

$$\frac{23{,}596 - 63{,}003}{63{,}003} \times 100 = -62.49\%. \tag{9}$$

However, suppose the retriever instead returns evidence for the wrong year:

- 2009 Revenue = \$54,132
- 2010 Revenue = \$63,003

The generator may then compute:

$$\frac{63{,}003 - 54{,}132}{54{,}132} \times 100 = 16.39\%. \tag{10}$$

This answer may be consistent with the retrieved evidence, but it does not answer the original question because the required 2011 value was not retrieved. The final numerical error therefore originates from the retrieval stage rather than from arithmetic computation alone. This distinguishes retrieval-induced failures from cases where the correct evidence is available but the model reasons incorrectly.

3.4.6 Abstention Despite Evidence

Abstention despite evidence occurs when the RAG system refuses to answer or states that the retrieved context is insufficient even though the necessary numerical evidence is present. For example, suppose the question asks:

> What is the percentage change in revenue from 2010 to 2011?

Suppose the retrieved evidence contains the required values:

- 2010 Revenue = \$63,003
- 2011 Revenue = \$23,596

The answer can be computed directly from the retrieved evidence:

$$\frac{23{,}596 - 63{,}003}{63{,}003} \times 100 = -62.49\%. \tag{11}$$

However, the model may output:

> The retrieved evidence is insufficient to answer the question.

In this case, the failure is not caused by missing retrieval evidence or by an incorrect arithmetic operation. Instead, the system fails to use evidence that is already available. This type of error is important because abstention is beneficial when the retrieved context genuinely lacks the information needed to answer the question, but unnecessary abstention reduces answer completeness when sufficient evidence is present.

In financial QA, abstention despite evidence may arise when the relevant numbers are retrieved but their surrounding context is fragmented or implicit. For instance, retrieved table fragments may include the required values while omitting the associated year, unit, company name, or row and column headers. The model may therefore treat the context as insufficient, even though the answer is recoverable from the retrieved evidence when the missing structural context is interpreted correctly.

3.5 Metrics

This thesis evaluates generated answers using both conventional answer-level metrics and numerical-faithfulness metrics. Existing financial RAG studies commonly emphasize retrieval effectiveness, answer-level quality, or LLM-based evaluation, but they do not always isolate numerical correctness, numerical grounding, and calculation-related reliability as separate dimensions. This thesis therefore focuses on three complementary aspects of numerical faithfulness: numerical correctness, numerical grounding, and retrieval-sensitive reliability. These empirical metrics approximate the conceptual conditions introduced in Section 3.2 by separately measuring numerical correctness, numerical grounding, and calculation-related reliability.

The metrics are designed to be lightweight and automatically computable from the generated answer, reference answer, and retrieved passages. They are implemented as a post-processing layer over the model outputs, allowing the same prediction files to be evaluated with both standard text-based metrics and numerical-faithfulness metrics. Rather than replacing existing RAG evaluation methods, these metrics are intended to make numerical failure patterns more visible in financial question answering, where an answer may be fluent and partially grounded but still numerically incorrect.

3.5.1 Numerical Correctness Metrics

Numerical correctness measures whether the generated answer contains a number that is close to the reference numerical answer.

Relative Numerical Error. Given a predicted numerical value n and a reference value n^*, we define relative numerical error as:

$$\text{RelErr}(n, n^*) = \frac{|n - n^*|}{\max(|n^*|, \epsilon)}, \tag{12}$$

where ϵ is a small constant used to avoid division by zero.

Because generated answers and reference answers may contain multiple numerical values, we compute the best relative error over all predicted-reference number pairs:

$$\text{BestRelErr}(A, A^*) = \min_{n \in \mathcal{N}(A),\, n^* \in \mathcal{N}(A^*)} \frac{|n - n^*|}{\max(|n^*|, \epsilon)}, \tag{13}$$

where $\mathcal{N}(A)$ denotes the numbers extracted from the generated answer and $\mathcal{N}(A^*)$ denotes the numbers extracted from the reference answer.

Numerical Match. A generated answer is considered numerically matched if at least one extracted predicted number is within a fixed relative-error tolerance of at least one extracted reference number:

$$\text{NumMatch}(A, A^*) = \mathbf{1}\left[\exists n \in \mathcal{N}(A),\, \exists n^* \in \mathcal{N}(A^*) \text{ such that } \text{RelErr}(n, n^*) < \tau\right]. \tag{14}$$

In our experiments, we use $\tau = 0.05$. This metric is less strict than exact string matching because it allows small numerical deviations while still penalizing large quantitative errors.

3.5.2 Numerical Grounding Metrics

Numerical grounding evaluates whether the numbers generated by the model appear in the retrieved evidence.

Numerical grounding in this thesis is narrower than full calculation grounding. It checks whether generated numbers are supported by the retrieved evidence, but it does not fully verify whether the model selected the intended operands or applied the correct operation. Calculation grounding would require checking the complete computation, including operand selection, operation choice, order of operations, units, and final numerical result. This would necessitate expensive symbolic analysis.

Direct Grounding Score. Let $\mathcal{N}(A)$ denote the numerical values extracted from the generated answer, and let $\mathcal{N}(\mathcal{R})$ denote the numerical values extracted from the retrieved passages. We define the direct grounding score as:

$$\text{DirectGrounding}(A, \mathcal{R}) = \frac{1}{|\mathcal{N}(A)|} \sum_{n \in \mathcal{N}(A)} \mathbf{1}[n \in \mathcal{N}(\mathcal{R})]. \tag{15}$$

This metric measures the fraction of generated numbers that directly appear in the retrieved context. If the generated answer contains no extracted numbers, the grounding score is undefined for that example.

Any Number Grounded. We also compute a binary grounding indicator:

$$\text{AnyGrounded}(A, \mathcal{R}) = \mathbf{1}\left[\exists n \in \mathcal{N}(A) \text{ such that } n \in \mathcal{N}(\mathcal{R})\right]. \tag{16}$$

This metric captures whether at least one generated numerical value is supported by the retrieved evidence.

These grounding metrics are intentionally conservative. They assess direct numerical overlap between extracted generated numbers and retrieved passages, but they do not prove that the generated answer was causally derived from the retrieved evidence or that the correct arithmetic operation was applied. For computed financial answers, the final numerical result may not appear explicitly in the retrieved passages even when the necessary operands are present.

Because the grounding metrics are based on direct numerical overlap, they may underestimate support for computed answers. In many financial QA examples, the final answer is derived from retrieved operands and does not appear explicitly in the retrieved passages. Therefore, a lower grounding score may reflect the limitation of direct overlap grounding rather than a complete absence of evidential support. This motivates reporting grounding separately from numerical match and analyzing lightweight verification as a complementary calculation-based signal.

4 Experiments and Analysis

4.1 Baseline Performance under Conventional and Numerical Metrics

Here, we evaluate the performance of the financial RAG configurations considered in this study using the two suites of evaluation metrics described in the previous chapter. First, we consider conventional text-based metrics, including exact match (EM), token-level F1, and a relaxed containment metric (i.e., whether the gold answer appears in the generated output). These metrics are then compared to the numerical faithfulness metrics developed in this thesis to highlight the differences between textual similarity and numerical faithfulness.

Experimental Setup. The experiments use 300 examples selected after shuffling the loaded FinQA examples with a fixed random seed. The subset is not manually filtered to include only numerical-answer questions, which keeps the selection procedure simple and reproducible. Among these 300 examples, 289 contain valid numerical ground-truth answers suitable for numerical-faithfulness evaluation.

Conventional text-based metrics, such as exact match, contains-gold, and token F1, are computed over all 300 generated outputs. Numerical metrics, including relative error, numerical match, and grounding-based measures, are reported over the 289 examples for which numerical evaluation is well-defined.

This subset supports the research problem by providing a controlled setting for studying numerical faithfulness under retrieval uncertainty. The goal is not to report full-benchmark state-of-the-art performance on FinQA, but to examine how numerical correctness, grounding, retrieval depth, preprocessing, and lightweight verification behave within the same financial RAG pipeline. Using a fixed subset allows the same examples to be compared across multiple retrieval configurations while keeping local generation cost manageable.

The subset is therefore suitable for the scope of this thesis, but it does not support broad claims about the full FinQA benchmark or all financial RAG systems. Future work could evaluate the framework on the full dataset, additional financial QA benchmarks, or a numerical-only subset constructed with explicit filtering criteria.

Retrieval Methods and Experimental Layout. The experiments evaluate three retrieval methods: BM25, keyword-overlap retrieval, and dense retrieval. BM25 and keyword-overlap retrieval are used throughout the baseline and preprocessing-ablation experiments because they are lightweight, interpretable sparse retrieval methods. BM25 serves as the main sparse lexical baseline, while keyword overlap is included as a simpler lexical comparison that scores passages by raw token overlap without inverse-document-frequency weighting or document-length normalization. Dense retrieval is added in Section 4.4 as an additional retrieval-model comparison under the revised linearized pipeline, using `sentence-transformers/all-MiniLM-L6-v2`.

The experimental chapter is organized in stages. Section 4.1 first reports the original baseline behavior for BM25 and keyword-overlap retrieval. Section 4.2 compares the original and improved preprocessing pipelines for the same two sparse retrieval methods. Section 4.4 then reuses the improved preprocessing setting and adds dense retrieval so that the revised sparse and dense retrieval methods can be compared under the same prompt, generator, top-K values, subset, and evaluation metrics. For this reason, some BM25 and keyword-overlap rows from the improved setting are repeated in Table 5: they are included as reference points for the dense-retrieval comparison rather than as a separate new ablation.

Implementation Details The retrieval models used in these experiments are BM25 and keyword-overlap retrieval. BM25 is the main sparse lexical retrieval baseline. It is implemented using the `rank_bm25` package, specifically `BM25Okapi`. Questions and passages are tokenized using a simple regular-expression tokenizer that lowercases text and extracts word tokens. BM25 scores are then computed between the tokenized question and the candidate passages, and duplicate passages are removed after ranking, and the top-K passages are passed to the generator.

The *Keyword* setting refers to a lightweight keyword-overlap retriever implemented in the project code. It tokenizes the question and each candidate passage using the same tokenizer as BM25, counts token frequencies, and scores each passage by the total overlapping token count between the question and passage. Unlike BM25, this retriever does not use inverse-document-frequency weighting or document-length normalization. It is included as a simple interpretable lexical baseline.

All experiments are implemented in Python. The main libraries used in the experimental pipeline include `rank_bm25` for BM25 retrieval, `numpy` for vector operations, `tqdm` for progress tracking, and `Ollama` for local language-model inference.

Generation is performed using `Qwen2.5:3B` through `Ollama`, with the exact model tag `qwen2.5:3b`. The pipeline calls the Ollama chat interface with the model name and a single user prompt containing the question and retrieved passages. No task-specific fine-tuning is performed. In the implementation used for these experiments, generation parameters such as temperature, top-p, maximum output length, and generation seed are not explicitly set in the project code; therefore, Ollama's default generation settings are used. The reported random seed, 42, is used only to shuffle the loaded FinQA examples before selecting the 300-example subset, not as a generation seed. Each configuration is evaluated using one generation run.

Unless otherwise stated, the experiments use the FinQA training split, a 300-example shuffled subset selected using random seed 42, retrieval depths $K \in \{3, 5, 8\}$, and one generation run per configuration. The main command-line parameters recorded by the pipeline include the dataset path, subset limit, retrieval method, retrieval depth, random seed (for shuffling), generator provider, generator model name, and output file name. The outputs are saved as JSONL records containing the question, gold answer, retrieved passages, retrieval scores, generated answer, retrieval configuration, model name, dataset path, and shuffle information.

The experiments were run in a Python virtual environment on the Computer Science Department server.

Evaluation Metrics. We report the following:

- **Exact Match (EM):** strict string-level equality between the generated and reference answers;
- **Contains Gold:** whether the reference answer appears as a substring of the generated answer;
- **Token F1:** token-level overlap between the generated and reference answers;
- **Relative Error (RelErr):** the best relative numerical error between extracted numbers in the generated answer and the reference answer;
- **Numerical Match (NumMatch):** whether at least one generated number matches a reference number within the predefined relative-error tolerance;
- **Grounding:** the fraction of generated numerical values that appear in the retrieved evidence;
- **Grounded:** whether at least one generated numerical value appears in the retrieved evidence.

For conventional answer-level metrics, this thesis reports scores over both the full generated output and the extracted `FINAL ANSWER` field when available. This allows surface-level answer quality to be compared under both full-response and final-answer-only evaluation.

For numerical-faithfulness metrics, the evaluator extracts predicted numbers from the generated output using the robust number extraction procedure. These extracted predicted numbers are then used for relative error, numerical match, direct grounding score, and any-number-grounded rate.

Therefore, the grounding metrics are based on numbers extracted from the generated output rather than on a separate reasoning-process annotation. Direct grounding measures whether extracted generated numbers appear in the retrieved passages, while any-number-grounded measures whether at least one extracted generated number appears in the retrieved passages.

Results. Table 2 summarizes the results. Relative error is reported using the median rather than the mean because mean relative error is highly sensitive to extreme numerical outliers. In financial QA, a small number of severe scale or extraction errors can make the mean relative error very large, while the median provides a more stable view of typical behavior.

The choice of median relative error affects how the numerical results should be interpreted. Mean and median relative error capture different aspects of numerical reliability. Mean relative error is sensitive to extreme outliers. This sensitivity can be useful because rare large numerical errors may be practically important in financial settings, especially when they reflect scale mistakes, unsupported numbers, or severe calculation failures. However, the mean can also be dominated by a small number of extreme cases and may not represent typical system behavior. Median relative error is therefore used in the main result tables to summarize typical numerical error, while mean relative error remains useful for identifying severe failure cases.

Token F1 should also be interpreted cautiously in this setting. It measures surface-level token overlap between the generated answer and the reference answer, but it does not directly evaluate whether the generated numerical value is correct, grounded in retrieved evidence, or produced through a valid calculation. For example, a response may share few tokens with the reference answer while still containing a numerically close value, or it may have higher textual overlap while using an incorrect number. For this reason, Token F1 is reported as a conventional comparison metric rather than as the main indicator of numerical faithfulness.

Table 2: Baseline financial RAG performance under BM25 and keyword-overlap retrieval on the same 300-example FinQA subset. Conventional metrics are computed over all 300 generated outputs, while numerical metrics are computed over the 289 examples with extractable numerical reference answers. Relative error is reported using the median to reduce sensitivity to extreme outliers.

Retriever	Top-K	EM	Contains	F1	Median RelErr	NumMatch	Grounding	Grounded
BM25	3	0.013	0.043	0.017	0.640	0.097	0.547	0.670
BM25	5	0.010	0.033	0.014	0.750	0.093	0.579	0.671
BM25	8	0.010	0.030	0.014	0.572	0.142	0.515	0.608
Keyword	3	0.010	0.043	0.019	0.767	0.100	0.548	0.641
Keyword	5	0.007	0.023	0.013	0.708	0.128	0.531	0.627
Keyword	8	0.003	0.027	0.006	0.546	0.145	0.506	0.592

Observations. Several patterns can be observed from Table 2. First, conventional text-based metrics are low across all configurations. Exact Match, Contains Gold, and Token F1 remain below the corresponding numerical match rates in most settings. This suggests that surface-level textual overlap may not fully describe numerical behavior in this experiment.

Second, increasing retrieval depth from Top-$K = 3$ to Top-$K = 8$ is associated with higher numerical match for both retrievers. BM25 increases from 0.097 to 0.142, while keyword overlap increases from 0.100 to 0.145. This suggests that retrieving more passages may expose the generator to more potentially useful numerical evidence in this setup.

Third, higher numerical match does not necessarily correspond to higher grounding. For BM25, the grounding score is highest at Top-$K = 5$ rather than Top-$K = 8$. For keyword overlap, the grounding score is highest at Top-$K = 3$ and decreases as Top-K increases. This indicates that adding more retrieved passages may also introduce additional numerical distractors or less directly relevant evidence.

Fourth, the comparison between BM25 and keyword overlap does not show one uniformly superior retriever. Keyword overlap gives slightly higher numerical match at Top-$K = 8$, while BM25 gives slightly stronger grounding at some smaller retrieval depths. These differences should be interpreted as behavior within this controlled setup rather than as a general conclusion about sparse retrieval methods.

Discussion. Overall, these results support the need to evaluate financial RAG using multiple numerical-faithfulness metrics rather than relying only on conventional answer-level metrics. In this experiment, text overlap, numerical match, median relative error, and grounding do not always move together. A configuration may produce more numerically close answers while also reducing grounding concentration, or it may produce grounded numbers without matching the final reference answer.

These results also suggest a retrieval-depth trade-off. Larger Top-K values may provide more opportunities for the generator to access relevant evidence, but they may also increase the amount of numerical context that the model must distinguish. Therefore, retrieval depth should not be interpreted as simply better or worse; it affects different dimensions of numerical faithfulness in different ways.

4.2 Ablation Study: Table Linearization and Year-Based Reranking

Our preliminary error analysis revealed that retrieval performance was sensitive to document preprocessing decisions, particularly table linearization and year-based reranking heuristics. In the original pipeline, financial tables were converted into textual passages using a simple linearization strategy, and retrieval scores could be adjusted using a year-overlap bonus. Manual inspection of retrieved passages identified two recurring issues:

- Financial tables in FinQA contain diverse layouts and hierarchical structures, causing naive linearization to sometimes weaken the association between row labels, column headers, units, and numerical values.

- Year-based reranking can over-prioritize passages containing years mentioned in the query, such as “2009” or “2010,” even when those passages do not contain the numerical evidence required to answer the question.

To investigate the consequences of these design choices, we compare the original pipeline with an improved preprocessing pipeline that:

1. uses a more robust table linearization strategy that detects unit rows, treats the first column as a row label when appropriate, and converts each table cell into a passage that preserves its row label, column header, unit, and table title when available; and

2. removes the year-based reranking bonus.

Experimental Setup. We compare the original pipeline against the improved preprocessing pipeline across both BM25 and keyword-based retrieval strategies. All other components of the pipeline, including the generator, prompt format, retrieval depths, and evaluation procedure, are kept fixed.

Results. Table 3 summarizes the results from the original and improved preprocessing pipelines.

Table 3: Ablation study comparing the original preprocessing pipeline with the improved table linearization and no-year-bonus pipeline on the same 300-example FinQA subset. Conventional metrics are computed over all 300 generated outputs, while numerical metrics are computed over the 289 examples with extractable numerical reference answers.

Setting	Retriever	Top-K	EM	Contains	F1	NumMatch	Median RelErr
Original	BM25	3	0.013	0.043	0.017	0.097	0.640
Improved	BM25	3	0.023	0.073	0.041	0.172	0.679
Original	BM25	5	0.010	0.033	0.014	0.093	0.750
Improved	BM25	5	0.017	0.087	0.035	0.189	0.563
Original	BM25	8	0.010	0.030	0.014	0.142	0.572
Improved	BM25	8	0.037	0.117	0.061	0.227	0.574
Original	Keyword	3	0.010	0.043	0.019	0.100	0.767
Improved	Keyword	3	0.013	0.067	0.022	0.172	0.563
Original	Keyword	5	0.007	0.023	0.013	0.128	0.708
Improved	Keyword	5	0.017	0.063	0.033	0.179	0.695
Original	Keyword	8	0.003	0.027	0.006	0.145	0.546
Improved	Keyword	8	0.010	0.063	0.029	0.192	0.555

Observations. The revised preprocessing pipeline improves several metrics in this experiment, including exact match, containment, token F1, and numerical match for the reported retrieval settings. The improvement in numerical match is especially visible for BM25. For example, BM25 Top-$K = 5$ increases from 0.093 to 0.189, and BM25 Top-$K = 8$ increases from 0.142 to 0.227.

These results suggest that preserving relationships among row labels, column headers, units, and numerical values may make retrieved passages more useful for downstream answer generation in this setup. In financial documents, the meaning of a number often depends on its surrounding table structure. Therefore, a preprocessing strategy that preserves more table context can help the model interpret retrieved numerical evidence more accurately.

The removal of the year-based reranking bonus may also contribute to the improvement. In the original pipeline, a passage could receive additional retrieval score simply because it mentioned the same year as the question. This heuristic may be unreliable when many nearby table rows, accounting categories, or reporting periods contain similar years. Removing this bonus reduces the risk of promoting passages that match the year but do not contain the required financial value.

Median relative error does not improve uniformly across all settings. In some configurations it decreases, while in others it remains similar or becomes slightly worse. This suggests that the revised preprocessing pipeline increases the frequency of numerically matched answers, but it does not remove all numerical or reasoning failures.

Discussion. The results from this ablation study indicate that preprocessing choices can affect numerical-faithfulness behavior in the evaluated financial RAG setup. This is expected because much of the relevant financial information appears in semi-structured tables rather than ordinary prose, and table structure can affect whether retrieved passages remain interpretable.

More broadly, this experiment suggests that numerical-faithfulness improvements do not necessarily require changing only the language model or retriever. In this setup, improving the representation of structured financial data also appears to improve several downstream metrics. This supports the importance of retrieval-aware preprocessing for financial RAG, while leaving broader validation across additional datasets and models as future work.

4.3 Numerical Failure Modes

This section analyzes the main types of numerical failures observed in the generated outputs. Instead of treating all incorrect answers as the same kind of error, we distinguish failures related to retrieval, grounding, answer construction, and numerical reasoning. This distinction is important because different errors require different remedies: one output may fail because the relevant evidence was not retrieved, while another may fail because the evidence was retrieved but the model selected the wrong value or applied the wrong operation.

The failure categories reported in this section are assigned using lightweight heuristics rather than manual validation. The heuristic procedure uses observable output patterns, including abstention phrases, missing numerical predictions, unsupported generated numbers, and mismatches between generated numbers and retrieved evidence. Therefore, the categories should be interpreted as heuristic labels rather than complete semantic explanations of the model's internal reasoning.

These heuristic labels are useful because they summarize likely sources of numerical failure in a transparent and interpretable way. However, they do not fully determine why the model produced a particular output. For example, an output labeled as an arithmetic or reasoning error may reflect an incorrect calculation, wrong operand selection, incomplete evidence, or another reasoning-related issue that cannot be fully separated by the current automatic procedure.

Future work should develop more explicit detection procedures for each failure category proposed in the taxonomy. Retrieval-induced errors, substitution errors, unit and scaling errors, hallucinated values, abstentions, and arithmetic errors could each be detected using more specialized rules, program-level checks, or human-validated annotations. Future work should also design separate metrics for the main components of numerical faithfulness, including numerical correctness, evidence grounding, and arithmetic consistency.

4.3.1 Operational Error Categorization

To analyze failure modes quantitatively, we apply an automatic categorization procedure to the evaluated model outputs. The procedure uses the generated answer, extracted predicted numbers, extracted reference numbers, numerical match, relative error, and direct grounding score.

The categorization is heuristic rather than a complete semantic error analysis. In particular, it does not fully determine whether the model selected the wrong operand, confused units, or applied the wrong arithmetic formula. Instead, it provides an interpretable approximation of common failure patterns observed in the generated outputs.

The automatic categories are defined as follows:

- **Abstention:** the generated answer contains phrases such as "insufficient evidence" or "cannot determine."
- **No Numerical Prediction:** no numerical value can be extracted from the generated output.
- **Hallucinated Number:** the output is numerically incorrect and has very low direct numerical grounding in the retrieved evidence.
- **Catastrophic Numerical Error:** the output is numerically incorrect and has an extremely large relative error, defined operationally as relative error greater than 100.

- **Grounded but Incorrect:** the output is numerically incorrect, but many of its extracted generated numbers appear in the retrieved evidence.
- **Correct but Weakly Grounded:** the output is numerically matched to the reference answer, but the extracted generated numbers have low direct grounding in the retrieved evidence.
- **Arithmetic / Reasoning Error:** a residual category for numerically incorrect outputs that are not classified as abstentions, missing numerical predictions, hallucinated numbers, or grounded-but-incorrect answers.

The heuristic rules are applied sequentially in the following implementation order. In this procedure, "very low direct numerical grounding" means a direct grounding score below 0.2, "low direct grounding" means a direct grounding score below 0.5, and "many extracted generated numbers appear in the retrieved evidence" means a direct grounding score above 0.5. In the implementation, an output is first checked for abstention phrases such as *insufficient evidence* or *cannot determine*. If no such phrase is found, the procedure checks whether any numerical prediction can be extracted. It then assigns *Hallucinated Number* when the output is numerically incorrect and the direct grounding score is below 0.2. It assigns *Grounded but Incorrect* when the output is numerically incorrect and the direct grounding score is above 0.5. It assigns *Catastrophic Numerical Error* when the relative error is greater than 100. It assigns *Correct but Weakly Grounded* when the output is numerically matched to the reference answer but the direct grounding score is below 0.5. All remaining outputs are assigned to the residual *Arithmetic / Reasoning Error* category.

This procedure allows us to summarize large numbers of outputs while preserving a distinction between unsupported numbers, grounded-but-wrong numbers, abstentions, and weakly grounded correct answers. These labels should be interpreted as automatic categories based on numerical overlap and grounding behavior rather than as manually verified explanations of the model's internal reasoning.

4.3.2 Empirical Error Distribution

We apply this categorization to outputs from the improved preprocessing pipeline using BM25 retrieval with Top-5 and Top-8 retrieved passages. Table 4 reports the resulting distribution of error categories.

Observations. No examples in the two BM25 settings reported in Table 4 were assigned to the *Catastrophic Numerical Error* category. The largest category in this analysis is abstention. For BM25 Top-$K = 5$, more than half of the evaluated outputs are categorized as abstentions. This high abstention rate may be related to the conservative prompt design and the fragmented structure

Table 4: Heuristic distribution of numerical failure modes under the improved preprocessing pipeline for BM25 retrieval. The distribution is computed over the 289 examples with extractable numerical reference answers from the 300-example FinQA subset.

Error Type	BM25 Top-5	BM25 Top-8
Abstention	54.64%	45.36%
Hallucinated Number	16.84%	23.71%
Correct but Weakly Grounded	12.03%	14.78%
Grounded but Incorrect	8.93%	8.25%
Arithmetic / Reasoning Error	6.87%	7.90%
No Numerical Prediction	0.69%	0.00%
Catastrophic Numerical Error	0.00%	0.00%

of retrieved financial evidence. Some retrieved passages may contain relevant numbers but lack enough surrounding context, such as the correct row label, year, unit, or table header. In such cases, the model may choose to answer "insufficient evidence" even when part of the needed evidence is present.

Increasing retrieval depth from Top-$K = 5$ to Top-$K = 8$ reduces the abstention rate from 54.64% to 45.36%. In this experiment, retrieving more passages appears to make the model more likely to produce a numerical answer. However, this should not be interpreted as a uniform improvement, because the change also affects other error categories.

The hallucinated-number category increases from 16.84% at Top-$K = 5$ to 23.71% at Top-$K = 8$. One possible explanation is that deeper retrieval exposes the generator to more candidate numerical values, including values that may be only weakly related to the question. This may increase the chance that the generated output includes a number that is not directly supported by the retrieved evidence.

The grounded-but-incorrect category remains present in both settings. These cases are important because they show that numerical overlap with retrieved evidence is not sufficient to guarantee a correct final answer. A model may reproduce numbers from the retrieved context while still selecting the wrong operand, using the wrong reporting period, or applying an incorrect calculation.

The correct-but-weakly-grounded category also appears in both settings. These cases suggest that numerical correctness and evidence grounding should be treated as separate dimensions. A generated answer may be close to the reference value even when the retrieved evidence does not strongly support the generated number according to the automatic grounding metric.

The arithmetic / reasoning-error category should be interpreted cautiously. In the automatic categorization used here, it functions as a residual category. It includes numerically incorrect outputs that are not classified as abstentions, missing numerical predictions, hallucinated numbers, or grounded-but-incorrect answers. Therefore, this label should not be interpreted as a fully verified explanation that the model made a specific arithmetic mistake. Instead, it indicates that the output remains incorrect after the simpler abstention- and grounding-based categories are excluded.

Discussion. This error distribution supports the need for evaluation beyond a single accuracy score. In the evaluated setting, incorrect or incomplete outputs arise in different ways: the model may abstain, generate an unsupported number, use a grounded number incorrectly, or produce a numerically close answer with weak grounding. These behaviors are not equivalent and may require different remedies.

The automatic categorization is not a substitute for manual error analysis. However, it provides a useful summary of likely failure patterns in the evaluated outputs. In particular, it shows that changing retrieval depth can shift the balance among abstention, hallucinated numbers, grounded-but-incorrect answers, and residual reasoning-related errors. This makes the evaluation more informative than reporting conventional answer-level metrics alone.

4.4 Additional Retrieval Model Comparison: Dense Retrieval

The previous sections compare BM25 and keyword-overlap retrieval under the original and improved preprocessing pipelines. To further examine whether the observed numerical-faithfulness patterns are specific to sparse lexical retrieval, this section adds a dense retrieval setting under the same revised linearized pipeline used in the improved rows of Table 3. This comparison is intended to isolate the retrieval-method dimension while keeping the dataset subset, preprocessing, generator, prompt format, top-K values, and evaluation metrics fixed.

The dense retriever is implemented using the `sentence-transformers` library. It uses the pretrained model `sentence-transformers/all-MiniLM-L6-v2` to encode questions and passages. Passage embeddings are computed once for the candidate passages of each example, and the question embedding is compared against them. Since embeddings are normalized in the implementation, the dot product is equivalent to cosine similarity. The top-K passages are then passed to the same generator and prompt template used by the sparse retrieval settings. The default sentence-transformer batch size is 32. The dataset, generator, prompt format, top-K values, and evaluation metrics are kept fixed so that the comparison focuses on the retrieval method. This experiment is not intended to provide an exhaustive benchmark of dense retrieval methods. Instead, it provides an additional retrieval-model comparison within the same controlled FinQA-based RAG setup. For consistency with the earlier experiments, numerical-faithfulness metrics are reported over the same 289 examples with valid numerical reference answers.

Table 5 repeats the improved BM25 and keyword-overlap rows from Table 3 so that the dense retrieval results can be compared directly against the sparse retrieval methods under the same revised linearized setting. The table is therefore not intended as a new preprocessing ablation; it is a retrieval-method comparison using the improved pipeline as the common setting.

Table 5: Retrieval-model comparison under the revised linearized setting on the same 300-example FinQA subset. Conventional metrics are computed over all 300 generated outputs, while numerical metrics are computed over the 289 examples with extractable numerical reference answers. Relative error is reported using the median to reduce sensitivity to extreme outliers.

Retriever	Top-K	EM	Contains	F1	Median RelErr	NumMatch	Grounding
BM25	3	0.023	0.073	0.041	0.679	0.172	0.499
BM25	5	0.017	0.087	0.035	0.563	0.189	0.505
BM25	8	0.037	0.117	0.061	0.574	0.227	0.440
Keyword	3	0.013	0.067	0.022	0.563	0.172	0.539
Keyword	5	0.017	0.063	0.033	0.695	0.179	0.496
Keyword	8	0.010	0.063	0.029	0.555	0.192	0.404
Dense	3	0.010	0.077	0.028	0.551	0.175	0.466
Dense	5	0.023	0.080	0.039	0.545	0.206	0.403
Dense	8	0.017	0.113	0.045	0.521	0.241	0.393

Table 5 compares BM25, keyword overlap, and dense retrieval under the revised linearized setting. Across the dense retrieval runs, increasing top-K from 3 to 8 increases the numerical match rate from 0.175 to 0.241. A similar pattern appears for BM25 and keyword overlap, where larger retrieval depth generally improves numerical match. This suggests that, in this experimental setup, retrieving more passages can increase the chance that the generated output contains a number close to the reference answer.

However, this improvement in numerical match does not imply uniformly better numerical faithfulness. For dense retrieval, the grounding score decreases from 0.466 at top-$K = 3$ to 0.393 at top-$K = 8$. BM25 and keyword overlap also show decreases in grounding at larger top-K values. This suggests that increasing retrieval depth may introduce additional numerical distractors or less directly useful evidence, even when it helps the model produce more numerical predictions.

Dense retrieval achieves the highest numerical match rate in this table at top-$K = 8$, but it does not achieve the highest grounding score. Conversely, keyword overlap at top-$K = 3$ has the highest grounding score among the revised linearized settings, but a lower numerical match rate. These results support the decision to report numerical correctness and grounding separately. A retriever can improve one dimension of performance without improving all dimensions of numerical faithfulness.

The dense retrieval comparison should be interpreted cautiously. The experiment uses one sentence-embedding retriever, one dataset subset, one generator, and the same lightweight evaluation metrics as the other experiments. Therefore, the results do not show that dense retrieval is generally better or worse than sparse retrieval for financial RAG. They only show that, under this controlled setup, the same trade-off between retrieval depth, numerical match, and grounding remains visible when an embedding-based retriever is added.

4.5 Lightweight Verification Strategies

Even when a RAG system retrieves useful financial evidence, the generated answer may still contain unsupported numbers or incorrect calculations. To analyze this issue, this thesis uses a lightweight post-generation verification procedure. The goal is not to correct model outputs or perform complete symbolic reasoning, but to flag whether a generated numerical answer is arithmetically consistent with numbers found in the retrieved evidence.

4.5.1 Post-Generation Percentage-Change Verification

The verifier focuses on a restricted but common class of financial questions involving percentage change, percent change, growth rate, increase, or decrease. These questions are suitable for lightweight checking because the relevant computation can often be expressed using two numerical operands.

Verification Procedure. For each generated answer, the verifier first determines whether the question contains one of the predefined percentage-change trigger phrases. If no trigger phrase is found, verification is not attempted.

For applicable questions, the verifier performs the following steps:

1. extract numerical values from the generated answer;
2. extract numerical values from the retrieved passages;
3. remove likely years, defined as values between 1900 and 2100;
4. treat the last extracted number in the generated answer as the predicted value;
5. enumerate ordered pairs of retrieved-context numbers as candidate operands; and
6. recompute percentage change for each candidate pair and compare it with the predicted value.

Given an old value a and a new value b, the verifier computes:

$$\text{PercentageChange}(a, b) = \frac{b - a}{|a|} \times 100. \tag{17}$$

For a predicted value $\hat{y}$, the verifier identifies the candidate pair whose recomputed percentage change has the smallest relative error with respect to $\hat{y}$:

$$\min_{a,b} \frac{|\hat{y} - \text{PercentageChange}(a, b)|}{|\hat{y}| + \epsilon}, \tag{18}$$

where ϵ is a small constant used for numerical stability. In implementation, the search is limited to the first 20 extracted context numbers to avoid excessive pair enumeration.

If the best recomputed value is within a fixed relative-error tolerance, the answer is marked as verified. Otherwise, it is marked as suspicious. The verifier does not overwrite or correct the generated answer; it only attaches a verification flag.

4.5.2 Verification Flags

The verification procedure assigns one of the following flags:

- **Not Applicable:** the question does not match the predefined percentage-change trigger patterns;
- **Verified:** the generated numerical value is consistent with at least one recomputed percentage change from retrieved-context numbers;
- **Suspicious:** verification is attempted, but no recomputed percentage change matches the generated value within tolerance;
- **No Predicted Number:** no numerical value is extracted from the generated answer;
- **Insufficient Context Numbers:** fewer than two numerical values are extracted from the retrieved passages;
- **No Valid Computation:** no valid percentage-change computation can be formed from the extracted context numbers.

These flags are intended for heuristics rather than automatic correction. A verified answer is arithmetically consistent with some pair of retrieved numbers, but this does not guarantee that the operands are semantically correct for the question. Conversely, a suspicious answer may reflect a genuine reasoning error, but it may also result from extraction noise, missing operands, or a question that requires reasoning beyond the verifier's supported pattern.

4.5.3 Impact on Numerical Faithfulness

We apply the verifier to outputs from the improved preprocessing pipeline using BM25 retrieval. Verification is attempted only for questions matching the predefined percentage-change-style triggers.

Verification Statistics. For BM25 Top-5, verification is attempted on 73 examples. Of these, 20 are marked as verified, corresponding to a verification rate of 27.4% among attempted cases. For BM25 Top-8, verification is again attempted on 73 examples, with 28 marked as verified, corresponding to a verification rate of 38.4%.

Retriever Setting	Attempted	Verified	Verification Rate
BM25 Top-5	73	20	27.4%
BM25 Top-8	73	28	38.4%

Table 6: Lightweight percentage-change verification results under the improved preprocessing pipeline. Verification rate is computed among examples where verification is attempted.

Observations. The verification results show that some generated answers can be arithmetically reproduced from numbers appearing in the retrieved context using the lightweight percentage-change verifier. In the evaluated setting, increasing retrieval depth from Top-$K = 5$ to Top-$K = 8$ increases the number of verified outputs. One possible explanation is that retrieving more passages exposes additional candidate operands that the verifier can use.

This result should be interpreted cautiously. Because the verifier enumerates candidate operand pairs from retrieved numbers, deeper retrieval also increases the number of possible numerical combinations. This can improve coverage, but it can also create spurious matches when unrelated numbers happen to produce a percentage change close to the generated answer. Therefore, a higher verification rate should not be treated as proof that the generated answer is semantically correct.

Discussion. The lightweight verifier is best understood as a heuristic signal rather than a complete correctness checker. It can identify generated outputs whose numerical values are consistent with at least one simple arithmetic operation over retrieved numbers. It can also flag outputs that are difficult to support using the retrieved numerical evidence. However, it does not fully determine whether the selected operands are the intended ones for the question.

In particular, the verifier does not fully resolve units, financial concepts, table headers, row labels, or reporting periods. These limitations are important because financial questions often depend on selecting the correct value from several similar numbers. As a result, an output may pass the lightweight verification check while still relying on the wrong semantic interpretation of the evidence.

Overall, these results suggest that lightweight post-generation verification can complement numerical-faithfulness metrics by adding an interpretable arithmetic-consistency check. At the same time, more reliable automatic verification or correction would require stronger operand selection, table understanding, and semantic interpretation than the simple verifier implemented in this thesis.

5 Conclusion and Future Work

5.1 Summary of Contributions

This thesis investigated the challenge of assessing numerical faithfulness in financial retrieval-augmented generation (RAG) systems. Although RAG pipelines can improve factual grounding by incorporating retrieved evidence, commonly used evaluation protocols remain heavily focused on textual similarity, answer overlap, and retrieval quality. These measures may fail to capture whether generated numerical outputs are quantitatively correct, properly grounded in retrieved evidence, or arithmetically consistent.

To address this limitation, this thesis introduced a structured framework for evaluating numerical faithfulness in financial RAG. Numerical faithfulness was defined in terms of numerical correctness, evidence grounding, and reasoning consistency. The thesis also proposed lightweight automatic metrics targeting numerical matching, relative error, numerical grounding, and robustness to retrieval variations.

Using a FinQA-based RAG setup over financial reports and structured tables, this thesis conducted an empirical analysis of lightweight financial RAG configurations under different retrieval depths, retrieval strategies, and preprocessing choices. The results showed that conventional textual metrics and numerical-faithfulness metrics can diverge substantially. In particular, stronger textual overlap or broader retrieval grounding did not necessarily imply more reliable numerical answers.

This thesis also developed a taxonomy of numerical failure modes, including retrieval-induced errors, arithmetic and computation errors, substitution errors, unit and scaling errors, hallucinated numerical values, and abstention despite relevant evidence. In addition, it explored lightweight post-generation verification strategies for analyzing arithmetic consistency in generated answers. Although the verification procedure could identify a subset of arithmetically supported outputs, it remained limited by ambiguous operand selection, semantic interpretation, and multi-step financial reasoning.

Overall, this work contributes a practical and interpretable framework for analyzing numerical reliability in financial RAG systems. The findings highlight the importance of evaluating grounded quantitative reasoning separately from conventional textual similarity and treating retrieval, answer construction, and numerical reasoning as distinct sources of failure.

5.2 Implications and Limitations

The results of this thesis have several implications for financial AI deployment. A response may be fluent, relevant, and partially grounded in retrieved evidence while still containing an incorrect final number. This distinction is important because many financial tasks depend on precise numerical interpretation, including percentage-change calculations, comparisons across reporting periods, and ratio-based reasoning.

The results also show that grounding and correctness should be treated as separate dimensions. A generated number may appear in the retrieved context but still be used incorrectly, selected from the wrong row or year, or combined with the wrong operation. Conversely, increasing retrieval depth can expose the model to more relevant evidence, but it can also introduce additional numerical distractors. This creates a trade-off between evidence coverage and context noise.

For deployment in financial settings, the evaluation framework would need stricter validation than the metrics used in this thesis. First, generated calculations should be checked against the expected calculation structure, including both the selected operands and the operations applied to them. Second, the system should identify the likely source of an error, such as wrong operand selection, incorrect operation choice, unit mismatch, or unsupported numerical generation. Finally, the source numbers used in the answer should be traceable to the retrieved evidence.

The experiments also show that conventional text-based metrics and numerical-faithfulness metrics should be interpreted as complementary rather than interchangeable. As discussed in Chapter 4, metrics such as Token F1 capture surface-level answer overlap, while numerical-faithfulness measures focus more directly on numerical correctness, grounding, and calculation-related reliability.

Despite these contributions, several limitations should be acknowledged. First, the main experiments use a 300-example subset of FinQA rather than the full benchmark. This setting is suitable for a controlled study of numerical correctness, relative error, grounding, and arithmetic consistency, but it does not support claims about full-benchmark state-of-the-art performance.

A related limitation is the computational scale of the experimental pipeline. The main bottleneck is not dataset loading, table linearization, number extraction, or metric computation, which are relatively lightweight once the FinQA examples are processed. Instead, the dominant cost comes from repeated local LLM generation. For each question and experimental configuration, the system must retrieve top-K evidence, construct a prompt, run the local model, extract the numerical answer, and evaluate the output.

This cost grows quickly when experiments are expanded to more examples, larger top-K values, additional retrieval models, preprocessing variants, verification settings, multiple random seeds, or larger generation models. Although local inference avoids hosted API costs and external rate limits, it is constrained by runtime, available hardware, model size, context length, and the number of configurations tested. Therefore, the experiments in this thesis should be interpreted as a controlled study of numerical-faithfulness failures rather than as a full-scale benchmark leaderboard.

Second, the experiments use a single generation run per configuration rather than multiple random seeds. Repeated runs would provide a better estimate of variance in numerical match, grounding, and abstention behavior.

Third, the failure-mode analysis is intended as an operational heuristic rather than a complete manual case-by-case annotation of every generated response. The dataset-level categorization relies on automatic signals such as extracted numbers, numerical match, relative error, grounding scores, and abstention behavior. This makes the analysis scalable and consistent across retrieval settings, but some categories should be interpreted as heuristic approximations rather than definitive semantic labels.

The failure categories reported in this thesis are assigned using lightweight heuristics rather than manual validation. These heuristics use observable output patterns, such as abstention phrases, missing numerical predictions, unsupported generated numbers, and mismatches between generated numbers and retrieved evidence. Therefore, the categories should be interpreted as intuitive labels rather than complete semantic explanations of the model's internal reasoning.

These heuristic labels are useful as user-facing information because they help summarize likely sources of numerical failure in a transparent and interpretable way. However, they do not fully determine why the model produced a particular output. For example, an output labeled as an arithmetic or reasoning error may reflect an incorrect calculation, wrong operand selection, incomplete evidence, or another reasoning-related issue that is not fully separable using the current automatic procedure.

Finally, the retrieval and verification methods explored in this thesis are intentionally lightweight. More advanced retrieval architectures, such as cross-encoder reranking, graph-based retrieval, table-aware retrieval, and agentic retrieval pipelines, were not extensively investigated. Similarly, the proposed verification mechanisms were limited in their ability to reliably identify relevant operands, perform multi-step reasoning, or resolve ambiguities in financial calculations.

5.3 Future Directions

This thesis identifies several focused directions for future work. First, future work could develop stronger methods for identifying the relevant operands and operations required by a financial question. A practical next step would be to extract candidate values together with their associated row labels, years, units, and financial concepts, then use this structured representation to constrain answer generation or verification.

Second, the lightweight verification procedure could be extended beyond simple percentage-change questions. Rather than enumerating a limited set of arithmetic expressions, future systems could identify the intended calculation from the question, construct a structured equation, and verify the generated answer against the retrieved evidence.

Third, retrieval and preprocessing could be optimized specifically for numerical evidence. Future work could investigate retrieval methods that explicitly consider financial variables, reporting periods, units, and table structure when selecting passages.

Future work should also develop more explicit implementations for each failure category proposed in the taxonomy. For example, retrieval-induced errors, substitution errors, unit and scaling errors, hallucinated values, abstentions, and arithmetic errors could each be detected using more specialized rules, program-level checks, or human-validated annotations.

Future work should also design separate metrics for each component of numerical faithfulness, including numerical correctness, evidence grounding, and arithmetic or calculation consistency. Such metrics would make it possible to evaluate not only whether the final answer is correct, but also whether the supporting numbers are retrieved, whether the correct operands are selected, and whether the calculation is valid.

Future larger-scale evaluation should account for computational scalability. Expanding the experiments to the full FinQA benchmark, additional financial QA datasets, multiple retrieval models, larger generation models, or repeated random seeds would increase the number of required retrieval-generation-evaluation runs. One practical approach is to first test on a smaller subset and then scale the most important configurations to the full benchmark.

Finally, the framework should be evaluated on a broader range of financial question types, including questions requiring both numerical reasoning and textual explanation, as well as multi-document and multi-period financial analysis. These directions would help move financial RAG systems from retrieving relevant passages toward producing answers whose numerical reasoning is transparent, verifiable, and reliable.

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